

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning & Mapping Exercise

COURSE NAME:	ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS	COURSE CODE:	MLA0101
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COURSE OUTCOME COVERED

CO3: *Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning. (BTL 3)*

Assessment Tool Weightage: 15%

Total Marks: 100

Time: 60 Mins

No. of Questions: 1

Annexure A

Logic Reasoning & Mapping Exercise

Code of Conduct:

I Yoga Madha A-192425058 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



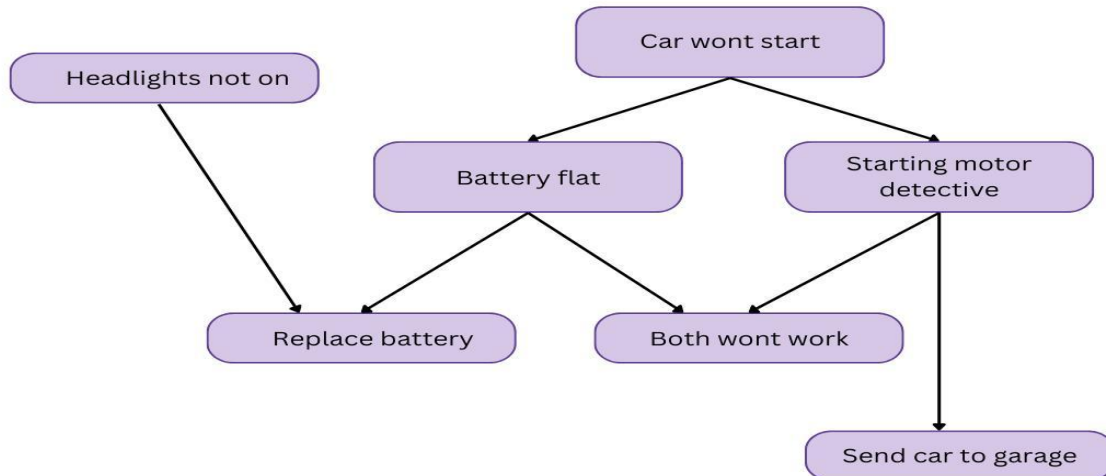
Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

Vehicle Fault Diagnosis

An automobile service center uses an expert system to diagnose vehicle faults. In this scenario, **Car101 does not start and its headlights are dim**. Based on these two symptoms, the expert system determines that Car101 may have a **battery problem**. A vehicle with a battery problem requires **battery inspection**, and vehicles requiring battery inspection are **marked for service**.

Related Diagram – Vehicle Fault Diagnosis



(i) Extract the Facts and Rules

Entities

The main entity in the scenario is:

Car101 – the vehicle being diagnosed.

Constant

The specific constant is:

Car101

In FOL, it can be written as:

car101

Predicates

Predicate	Meaning
DoesNotStart(x)	Vehicle x does not start
DimHeadlights(x)	Vehicle x has dim headlights
BatteryProblem(x)	Vehicle x has a battery problem
BatteryInspection(x)	Vehicle x requires battery inspection
MarkedForService(x)	Vehicle x is marked for service

Facts

Fact 1:

DoesNotStart(Car101)

Fact 2:

DimHeadlights(Car101)

Rules

Rule 1:

Vehicles that do not start and have dim headlights may have a battery problem.

DoesNotStart(x) $\wedge$ DimHeadlights(x) $\rightarrow$ BatteryProblem(x)

Rule 2:

Vehicles with a battery problem require battery inspection.

BatteryProblem(x) → BatteryInspection(x)

Rule 3:

Vehicles requiring battery inspection are marked for service.

BatteryInspection(x) → MarkedForService(x)

(ii) Propositional Logic Representation

Let:

- **P** = Car101 does not start.
- **Q** = Car101 has dim headlights.
- **R** = Car101 has a battery problem.
- **S** = Car101 requires battery inspection.
- **T** = Car101 is marked for service.

Facts

P

Q

Therefore:

P ∧ Q

Rule 1

(P ∧ Q) → R

This means:

If Car101 does not start AND has dim headlights, then Car101 has a battery problem.

Rule 2

R → S

This means:

If Car101 has a battery problem, then Car101 requires battery inspection.

Rule 3

S → T

This means:

If Car101 requires battery inspection, then Car101 is marked for service.

Complete Propositional Representation

P ∧ Q

(P ∧ Q) → R

R → S

S → T

Therefore:

P ∧ Q → R → S → T

(iii) First-Order Logic Representation

First-Order Logic represents the general rules using **predicates, variables and quantifiers**.

Predicates

DoesNotStart(x)

DimHeadlights(x)

BatteryProblem(x)

BatteryInspection(x)

MarkedForService(x)

Here, x represents any vehicle.

Facts

DoesNotStart(Car101)

DimHeadlights(Car101)

Rule 1

For every vehicle, if it does not start and has dim headlights, then it has a battery problem.

$\forall x [(DoesNotStart(x) \wedge DimHeadlights(x)) \rightarrow BatteryProblem(x)]$

Rule 2

For every vehicle, if it has a battery problem, then it requires battery inspection.

$\forall x [BatteryProblem(x) \rightarrow BatteryInspection(x)]$

Rule 3

For every vehicle, if it requires battery inspection, then it is marked for service.

$\forall x [BatteryInspection(x) \rightarrow MarkedForService(x)]$

Complete FOL Knowledge Base

1. **DoesNotStart(Car101)**

2. **DimHeadlights(Car101)**

3. **$\forall x [(DoesNotStart(x) \wedge DimHeadlights(x)) \rightarrow BatteryProblem(x)]$**

4. **$\forall x [BatteryProblem(x) \rightarrow BatteryInspection(x)]$**

5. **$\forall x [BatteryInspection(x) \rightarrow MarkedForService(x)]$**

Predicates, Variables, Constants and Quantifiers

Component	Representation	Meaning
Variable	x	Any vehicle
Constant	Car101	Specific vehicle
Universal Quantifier	$\forall x$	For every vehicle x
AND	$\wedge$	Both conditions are true
Implication	$\rightarrow$	If...then...

The **universal quantifier $\forall x$** is used because the rules are applicable to every vehicle.

(iv) Unification and Substitution

Unification finds a substitution that makes two predicates identical.

Consider:

DoesNotStart(x)

and

DoesNotStart(Car101)

The required substitution is:

$\theta = \{x / Car101\}$

Therefore:

DoesNotStart(x)

becomes:

DoesNotStart(Car101)

Similarly:

DimHeadlights(x)

and

DimHeadlights(Car101)

can be unified using:

$\theta = \{x / \text{Car101}\}$

Therefore:

DimHeadlights(Car101)

Now substitute $x = \text{Car101}$ into Rule 1:

DoesNotStart(x) $\wedge$ DimHeadlights(x) $\rightarrow$ BatteryProblem(x)

becomes:

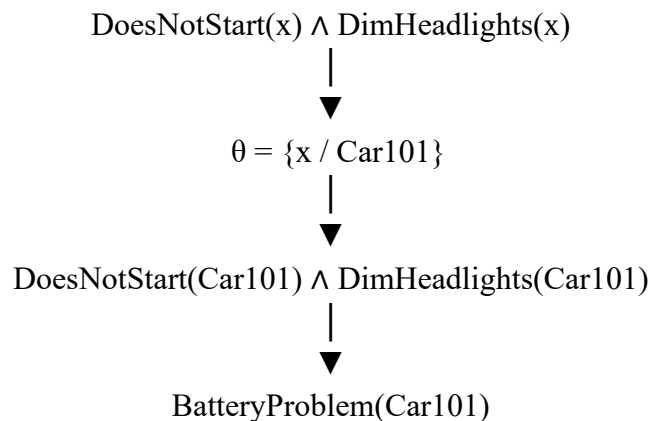
DoesNotStart(Car101) $\wedge$ DimHeadlights(Car101) $\rightarrow$ BatteryProblem(Car101)

Since both conditions are true:

BatteryProblem(Car101)

Unification Diagram

General Rule:



(v) Resolution to Determine Battery Problem

We want to prove:

BatteryProblem(Car101)

Resolution works by converting the rule into clause form.

Step 1: Original Rule

$\forall x [(\text{DoesNotStart}(x) \wedge \text{DimHeadlights}(x)) \rightarrow \text{BatteryProblem}(x)]$

Remove the implication:

$\neg(\text{DoesNotStart}(x) \wedge \text{DimHeadlights}(x)) \vee \text{BatteryProblem}(x)$

Using De Morgan's Law:

$\neg\text{DoesNotStart}(x) \vee \neg\text{DimHeadlights}(x) \vee \text{BatteryProblem}(x)$

Therefore, the clause is:

$\neg\text{DoesNotStart}(x) \vee \neg\text{DimHeadlights}(x) \vee \text{BatteryProblem}(x)$

Step 2: Substitute Car101

Using:

$\theta = \{x / \text{Car101}\}$

we get:

$\neg \text{DoesNotStart}(\text{Car101}) \vee \neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$

Step 3: Use Fact 1

Fact:

$\text{DoesNotStart}(\text{Car101})$

Resolve:

$\neg \text{DoesNotStart}(\text{Car101}) \vee \neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$

with:

$\text{DoesNotStart}(\text{Car101})$

The result is:

$\neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$

Step 4: Use Fact 2

Fact:

$\text{DimHeadlights}(\text{Car101})$

Resolve:

$\neg \text{DimHeadlights}(\text{Car101}) \vee \text{BatteryProblem}(\text{Car101})$

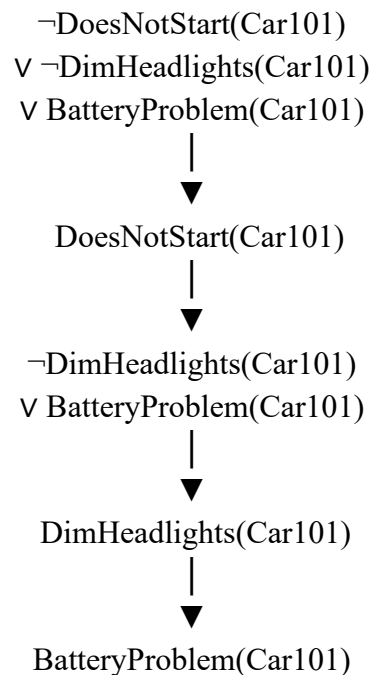
with:

$\text{DimHeadlights}(\text{Car101})$

Therefore:

$\text{BatteryProblem}(\text{Car101})$

Resolution Flow

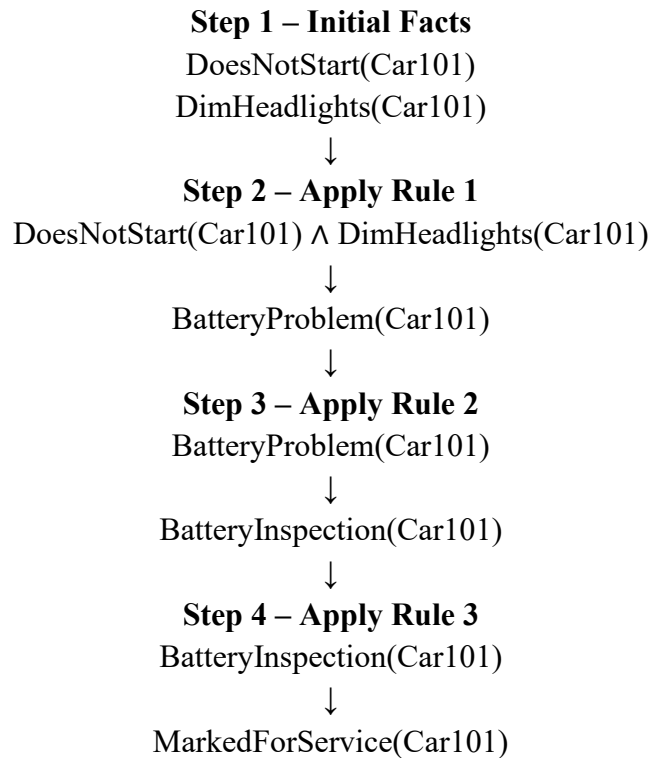


Therefore:

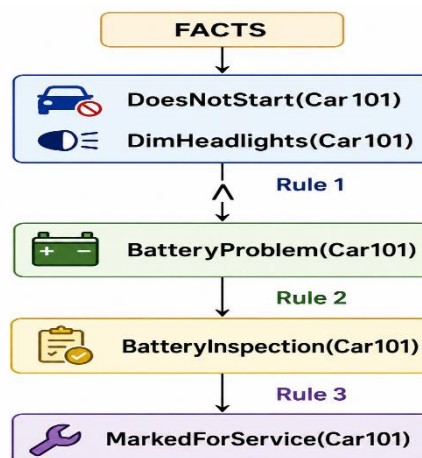
$\therefore \text{BatteryProblem}(\text{Car101})$

(vi) Complete Inference Sequence

Once the battery problem is established, the remaining rules can be applied.



Complete Knowledge Inference Diagram



Conclusion

Therefore, Car101 has a battery problem. Since a vehicle with a battery problem requires battery inspection, Car101 requires battery inspection. Since vehicles requiring battery inspection are marked for service, Car101 should be marked for service.

Final Inference

DoesNotStart(Car101) $\wedge$ DimHeadlights(Car101)

→ BatteryProblem(Car101)

→ BatteryInspection(Car101)

→ MarkedForService(Car101)

Hence, the AI expert system correctly concludes that Car101 has a battery problem and should be marked for service.

EVALUATION RUBRICS:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
Problem Understanding & Knowledge Identification	15%	13–15% – Clearly identifies entities, facts, relationships, assumptions, and reasoning requirements from the given scenario.	10–12% – Identifies most relevant knowledge with minor omissions.	7–9% – Identifies basic knowledge but misses several important elements.	0–6% – Poor or incorrect identification of the given knowledge.
Logical Representation & Mapping	20%	17–20% – Accurately maps real-world statements into appropriate propositional and first-order logic representations.	13–16% – Provides mostly correct logical mappings with minor errors.	9–12% – Provides partial mappings with several logical errors.	0–8% – Logical mapping is largely incorrect or missing.
Predicates, Variables, Constants	15%	13–15% – Correctly identifies and applies predicates, variables, constants, universal and existential quantifiers.	10–12% – Uses the concepts correctly with minor errors.	7–9% – Demonstrates partial understanding with some incorrect usage.	0–6% – Concepts are incorrectly applied or missing.
Logical Connectives	10%	9–10% – Correctly applies AND, OR, NOT, implication, and equivalence to	7–8% – Mostly correct use of connectives with minor errors.	5–6% – Basic use of connectives but several errors occur.	0–4% – Incorrect or incomplete formula construction.